IC3 GS5

Computing Fundamentals

Student Workbook Handouts

Name: _____

Class / Program: _____

Instructor: Jmal Chestnut

How to use these handouts

- Bring this packet to every session. Each unit has a **notes guide** (fill in the blanks during the lesson), **key vocabulary**, **practice questions**, and an **'I can...' checklist.**
- Use your flashcards every day for 5 minutes. The exam uses the vocabulary words.
- Track your GMetrix practice scores on the GMetrix Progress Tracker at the back. Goal: **85% or higher in Testing Mode.**
- Practice with made-up information only. Never type real passwords or personal documents on class computers.

UNIT 1 · SESSION 1

Hardware Basics: Devices, Memory, Peripherals & Power

Name: _____ Date: _____

Notes Guide

1. RAM is _____ memory: it is erased when the power is off. Storage is _____.
2. A _____ accepts requests from other computers on a network.
3. HDMI carries _____ and _____. VGA carries only _____.
4. External speakers usually connect with _____ or a _____ mm audio jack.
5. A _____ lets the operating system talk to hardware such as a printer.
6. Sleep vs. Hibernate: _____
7. One pro and one con of touchscreens: _____

Key Vocabulary

Term	What it means
Hardware	The physical parts of a computer or device you can touch.
Server	A computer or program that accepts requests from other computers (clients) on a network and provides services or data back to them.
Laptop	A portable computer with a built-in screen, keyboard, touchpad, and battery.
PC	Personal Computer. Often used to mean a computer running Windows (as opposed to a Mac).
Memory	Usually means RAM: the computer's short-term working space. It is volatile, so it's erased when power is off. More RAM lets you run more programs at once. (Long-term storage is the drive.)
Peripherals	External devices you connect to a computer to add input, output, or storage.
USB	Universal Serial Bus. The standard port for connecting and charging devices. USB-A is the older rectangle; USB-C is the newer small oval that plugs in either way.
HDMI	High-Definition Multimedia Interface. A digital port and cable that carries HD video AND audio together.
VGA	Video Graphics Array. An older analog video-only port (usually blue, 15 pins) for monitors and projectors. It does not carry audio.
Driver	Software that lets the operating system talk to a piece of hardware, like a printer or graphics card.
Mouse	A handheld pointing device used to move the pointer and click on the screen.
Keyboard Shortcut	A combination of keys that runs a command without using the mouse.

UNIT 1: Hardware Basics: Devices, Memory, Peripherals & Power (continued)

Practice Questions

1. Name three peripherals and say whether each one is input, output, or storage.

2. Your phone says 'Storage Full.' List two things you could do.

3. Write the steps to set your PC to sleep after 30 minutes when plugged in.

4. Why should you download drivers only from the manufacturer or Windows Update?

5. You're editing a 2-page resume. Would you choose a phone, tablet, or laptop? Why?

I can... (check when you're confident)

- ☐ Identify types of devices: server, laptop, desktop (plus tablets, all-in-ones, 2-in-1s) (1.2.a)
- ☐ Know the impact of memory (RAM) and storage on usage (1.2.b)
- ☐ Connect peripherals: camera, audio, microphone, printer, USB devices, external display (1.2.c)
- ☐ Understand power management and power settings (1.2.f)
- ☐ Understand what drivers do and whether a driver is compatible with a device (1.2.g)
- ☐ Understand common hardware configurations (1.2.l)
- ☐ Know the implications of device choice for document usage (1.2.m)
- ☐ Weigh the pros and cons of touchscreen vs. non-touchscreen devices (1.2.n)

UNIT 2 · SESSION 2

Internet & Networking Basics

Name: _____ Date: _____

Notes Guide

1. The three main ways to connect: _____, _____, and _____.
2. A modem translates _____ signals into _____ information.
3. A router transfers data from one _____ to another.
4. Bandwidth is purchased through an _____ (Internet Service Provider).
5. The most common wired connection uses an _____ cable (RJ-45).
6. IPv4 example: _____ IPv6 example: _____
7. To find my IP address I can type _____ in Command Prompt.

Key Vocabulary

Term	What it means
Modem	The device that connects your home to your Internet Service Provider (ISP). It converts the ISP's signal (cable, DSL, fiber) into digital data your devices can use.
Router	A device that directs data between networks, such as your home network and the Internet. Most home routers also create the Wi-Fi network.
Bandwidth	In networking, the maximum amount of data a connection can move in a set amount of time, usually measured in Mbps or Gbps. More bandwidth = more data at once.
Ethernet	The standard for wired networks. Uses network cables with RJ-45 plugs (they look like a wide phone plug).
Wi-Fi	Wireless networking technology that connects devices to a router using radio waves.
Wireless	Connecting without cables. Includes Wi-Fi, Bluetooth, and cellular.
Cellular Network	A network of cell towers, each covering an area called a 'cell,' that carriers use for calls, texts, and mobile data.
IPv4	Internet Protocol version 4. Uses 32-bit addresses, which allows about 4.3 billion addresses. That's not enough for today's devices.
IPv4 Address	An address written as four decimal numbers from 0 to 255, separated by dots.
IPv6	Internet Protocol version 6. Uses 128-bit addresses, which allows a practically unlimited number. It was created because IPv4 addresses ran out.
IPv6 Address	An address written as eight groups of four hexadecimal digits (0-9, a-f), separated by colons. Groups of zeros can be shortened with ::.
Secure Network	A network that requires a password or sign-in and encrypts traffic (for example, WPA2 or WPA3 Wi-Fi).

UNIT 2: Internet & Networking Basics (continued)

Practice Questions

1. Compare cellular, Wi-Fi, and wired networks: list one pro and one con each.

2. Write the steps to join the network 'Library-Guest' with the password Read2Day!

3. Your Wi-Fi is slow at night when everyone is home. Explain why, using the word bandwidth.

4. Circle the valid IPv4 addresses: 10.0.0.5, 192.168.1.300, 172.16.4.20, fe80::1

5. On a public network, should you click Yes or No to 'Allow your PC to be discoverable'? Why?

I can... (check when you're confident)

- ☐ Understand the use of Ethernet ports (1.2.d)
- ☐ Connect a device to a wireless (Wi-Fi) network (1.2.e)
- ☐ Know the difference between cellular, Wi-Fi, and wired networks (1.2.j)
- ☐ Understand Internet connection concepts: modem, bandwidth/speed, router, wired, wireless (1.2.k)
- ☐ Define IP address (1.3.k)
- ☐ Know how to find an IP address on a personal computer (1.3.m.iv)
- ☐ Know how to determine a connection speed (1.3.m.v)

UNIT 3 · SESSION 3

Mobile Devices

Name: _____ Date: _____

Notes Guide

1. Cell phones connect through _____ that each cover an area called a cell.
2. A cellular tablet can get online but usually can't be used as a _____.
3. Apple's mobile OS is _____. Google's is _____.
4. Voicemail tips: be clear, concise, and use an appropriate _____.
5. SMS = _____ only. MMS = _____, _____, group texts.
6. Two common ways to manage interruptions: _____ and alarms / Do Not Disturb.
7. Prepaid vs. postpaid: _____

Key Vocabulary

Term	What it means
Cellular Network	A network of cell towers, each covering an area called a 'cell,' that carriers use for calls, texts, and mobile data.
Cellular Data	Internet access through your phone carrier's cell network. It usually counts against your plan's data.
iOS	Apple's operating system for the iPhone. (iPads use iPadOS, which is closely related.)
Android	Google's mobile operating system, built on Linux and open source. Used by Samsung, Google Pixel, Motorola, and many other brands.
SMS	Short Message Service. Plain text messages of up to 160 characters sent over the cellular network.
MMS	Multimedia Messaging Service. Text messages that include pictures, video, audio, or group texts.
Text Messaging	Sending short written messages between phones or devices (SMS, MMS, iMessage, RCS).
VoIP	Voice over Internet Protocol. Making phone or video calls over the Internet instead of the regular phone network.
Wireless	Connecting without cables. Includes Wi-Fi, Bluetooth, and cellular.
Permissions	Settings that control who can view, change, or delete a file or folder, or what an app is allowed to use (camera, location, contacts).

UNIT 3: Mobile Devices (continued)

Practice Questions

1. Explain the difference between prepaid and postpaid phone plans. Which might be better for someone rebuilding credit?

2. Write a professional 15-second voicemail greeting.

3. List the steps to stop a game app from sending notifications on your phone.

4. Give one reason a business might still use hardwired desk phones.

5. Rewrite this text to a supervisor professionally: 'yo im running l8 traffic'

I can... (check when you're confident)

- ☐ Understand cellular phone concepts (1.1.a)
- ☐ Be familiar with cellular-enabled tablets (1.1.b)
- ☐ Be familiar with smartphones (1.1.c)
- ☐ Understand the use of hardwired phones (1.1.d)
- ☐ Know how to configure and use voicemail (1.1.e)
- ☐ Use instant messaging (1.1.f)
- ☐ Know how to configure notifications (1.1.g)
- ☐ Platform considerations: iOS and Android (1.2.h.iv-v)

UNIT 4 • SESSION 4

Operating Systems & Software Architecture

Name: _____ Date: _____

Notes Guide

1. An operating system manages all of a device's installed _____ and _____.
2. Linux is an _____ operating system available to the public.
3. Customize Windows in Control Panel under _____ and Personalization.
4. Install, uninstall, and update software in Windows using _____ (or Settings > Apps).
5. Radio button = choose _____. Check box = choose _____.
6. Title bar buttons: _____, _____, _____.
7. Administrator vs. Standard account: _____

Key Vocabulary

Term	What it means
OS	Operating System. The main software that manages all the hardware and software on a device and lets you use it.
Linux	A free, open-source operating system that anyone can use and change. Comes in many versions (distributions) such as Ubuntu. Runs most web servers.
Firmware	Software stored permanently on a hardware device that controls its basic functions (for example, BIOS/UEFI on a PC or the software inside a router).
Radio Button	A small round option in a group. You can pick only ONE choice from the group.
Check Box	A small square option in a menu or form. You can select as many check boxes as you want.
Drop-Down Menu	A list that opens when you click it, letting you choose one option.
Keyboard Shortcut	A combination of keys that runs a command without using the mouse.
Permissions	Settings that control who can view, change, or delete a file or folder, or what an app is allowed to use (camera, location, contacts).
Local App	An app installed on your device. It runs from your device, often works offline, and uses your device's storage.
.exe	An executable file: a Windows program that runs or installs software when you open it.

UNIT 4: Operating Systems & Software Architecture (continued)

Practice Questions

1. Compare Windows, macOS, and Linux: one similarity and one difference.

2. Write the steps to set a screen saver to start after 10 minutes.

3. Give an example of an application setting and a global setting.

4. Why should you download software only from official sites or app stores?

5. Explain why a business would NOT give every employee administrator rights.

I can... (check when you're confident)

- ☐ Know platform implications: Mac, PC, Linux, iOS, Android, firmware (1.2.h)
- ☐ Understand OS versioning and update awareness (1.3.a)
- ☐ Know application vs. operating system vs. global settings (1.3.b)
- ☐ Understand OS/software settings: desktop customization; window management (minimize, maximize, resize) (1.3.c)
- ☐ Set software preferences (1.3.d)
- ☐ Users and profiles (1.3.e)
- ☐ Menu navigation: radio buttons, check boxes, drop-down menus, multi-select (1.3.h)
- ☐ Rights and permissions (administrative rights) (1.3.j)
- ☐ Install, uninstall, update, and repair software, from the Internet and from local media (DVD-ROM, USB) (1.3.l)

UNIT 5 · SESSION 5

File Management, File Types & File Sharing

Name: _____ Date: _____

Notes Guide

1. The _____ file extension is produced by Adobe Flash.
2. _____ apps typically organize items (eBooks, songs, videos) into folders/libraries.
3. Scanning is the process of _____ a physical item into a digital medium.
4. .docx = _____ .xlsx = _____ .pdf = _____ .exe = _____
5. To zip: right-click > _____ > Compressed (zipped) folder.
6. Read-only is set in _____ > General tab.
7. Shared link vs. shared folder: _____

Key Vocabulary

Term	What it means
.doc	The older Microsoft Word document format (Word 97-2003).
.docx	The current Microsoft Word document format (Word 2007 and newer). Smaller and more modern than .doc.
.txt	A plain text file. Just letters and numbers, no bold, colors, or pictures.
.xls	The older Microsoft Excel spreadsheet format (Excel 97-2003).
.xlsx	The current Microsoft Excel spreadsheet format (Excel 2007 and newer).
.exe	An executable file: a Windows program that runs or installs software when you open it.
.swf	Shockwave Flash file: Adobe Flash animation or game. Flash was shut down at the end of 2020, so browsers no longer play .swf files.
.pdf	Portable Document Format. Looks exactly the same on any device, so it's great for forms and resumes.
.wav	An uncompressed audio file. High quality but much larger than .mp3.
.zip	A compressed archive file that packs one or more files into one smaller file.
.jpg	JPEG image. A compressed photo format that supports 24-bit color (about 16.7 million colors). It shrinks files by dropping some detail (lossy).
.tif	Tagged Image File Format (.tif or .tiff). A high-quality image format, often used for scanning and printing. Files are large.
.mp3	A compressed audio file. Small enough to store thousands of songs.
.mp4	A very common video file format (video + audio) that plays on almost any device.
.avi	Audio Video Interleave. An older Microsoft video file format that holds video and sound together.
Zip File	A compressed file (.zip) that bundles one or more files into a smaller package for easier sharing or storage. You 'extract' (unzip) it to use the files.
Scanning	Turning a paper document or photo into a digital file using a scanner or a phone camera app. (Also: an antivirus 'scan' checks files for malware.)
Permissions	Settings that control who can view, change, or delete a file or folder, or what an app is allowed to use (camera, location, contacts).

UNIT 5: File Management, File Types & File Sharing (continued)

Practice Questions

1. Match each extension to the kind of file: .mp4, .jpg, .txt, .zip, .xlsx, .wav

2. List the steps to rename a folder on your desktop.

3. You have 12 scanned pages to send a caseworker or counselor. What's the best way to send them and why?

4. What's the difference between moving and copying a file?

5. Name two apps you could use to manage eBooks, music, or videos.

I can... (check when you're confident)

- ☐ Move and rename files and folders (1.3.f.i)
- ☐ File permissions (1.3.f.ii)
- ☐ File extensions: .doc/.docx, .txt, .xls/.xlsx, .exe, .swf, .pdf, .wav, .zip, .jpg, .tif, .mp3, .mp4, .avi (1.3.f.iii)
- ☐ Manage electronic media: eBooks, songs, videos (1.3.f.iv)
- ☐ Document management: scanning documents, taking pictures (1.3.g)
- ☐ Search for files (1.3.i)
- ☐ File transfer options: attach files to email; publicly shared vs. shared links vs. shared folders (1.5.a)
- ☐ Understand the need for file compression (zipping) when sharing (1.5.b)

UNIT 6 · SESSION 6

Troubleshooting, Compatibility, Backup & Restore

Name: _____ Date: _____

Notes Guide

1. Two signs of malware: receiving an _____ or when an app _____.

2. A device may be limited if it can only connect to the Internet _____.

3. A _____ is software that adds functionality to an application.

4. My troubleshooting steps: 1) _____ 2) _____ 3) _____ 4) _____

5. Hardware clue: _____ Software clue: _____

6. 3-2-1 backup rule: ____ copies, ____ types of storage, ____ off-site.

7. Reset this PC options: Keep my files vs. _____

Key Vocabulary

Term	What it means
Troubleshooting	A step-by-step process for figuring out what's wrong and fixing it.
Hardware	The physical parts of a computer or device you can touch.
Driver	Software that lets the operating system talk to a piece of hardware, like a printer or graphics card.
Malware	Malicious software. The umbrella term for any software designed to harm, spy on, or take control of a device (viruses, Trojans, ransomware, spyware).
Cloud Storage	Saving files on a company's remote servers (Google Drive, OneDrive, iCloud, Dropbox) instead of only on your own device.
Firmware	Software stored permanently on a hardware device that controls its basic functions (for example, BIOS/UEFI on a PC or the software inside a router).

UNIT 6: Troubleshooting, Compatibility, Backup & Restore (continued)

Practice Questions

1. Your computer froze while typing an essay. List what you would do, in order.

2. Decide hardware or software: (a) one app crashes after an update, (b) the monitor shows nothing and its power light is off, (c) the printer isn't detected.

3. Write a clear help-desk message about a problem you've had.

4. Design a backup plan for your phone photos.

5. What's the difference between 'Keep my files' and 'Remove everything'?

I can... (check when you're confident)

- ☐ Know basic problem-solving techniques (1.3.m.i)
- ☐ Determine problems involving hardware vs. software (1.3.m.ii)
- ☐ Use proper terminology to describe the problem (1.3.m.iii)
- ☐ Know platform compatibility, including media compatibility issues ('Why won't my video play?') (1.2.i)
- ☐ Understand backup concepts (1.4.a)
- ☐ Know how to back up and restore (1.4.b)
- ☐ Know how to complete a full system restore on a personal device (1.4.c)

UNIT 7 · SESSION 7

Cloud Computing

Name: _____ Date: _____

Notes Guide

1. Cloud storage can back up data on a _____ basis and be accessed anywhere with _____.
2. Cloud services are mostly free, but payment is required for _____.
3. A _____ is used to administer, deliver, and track progress in eLearning.
4. Google Docs, Sheets, and Slides are used in _____.
5. Cloud storage vs. cloud access: _____
6. Web app vs. local app: _____
7. Three online storage apps: _____, _____, _____

Key Vocabulary

Term	What it means
Cloud Storage	Saving files on a company's remote servers (Google Drive, OneDrive, iCloud, Dropbox) instead of only on your own device.
Cloud Access	Using apps, files, and services over the Internet from any device, instead of from one computer.
SaaS	Software as a Service. Software you use over the Internet, usually by subscription. The company hosts it, updates it, and stores the data.
LMS	Learning Management System. Software that delivers online courses, gives assignments and tests, and tracks student progress.
Web App	An app that runs inside a web browser, with nothing to install. Needs an Internet connection (mostly).
Local App	An app installed on your device. It runs from your device, often works offline, and uses your device's storage.
Server	A computer or program that accepts requests from other computers (clients) on a network and provides services or data back to them.

UNIT 7: Cloud Computing (continued)

Practice Questions

1. Define cloud storage and cloud access in your own words.

2. List one benefit each for sharing, collaborating, and storing in the cloud.

3. What does SaaS mean? Give two examples.

4. Write the steps to upload a file to Google Drive and share it as Viewer.

5. Name an LMS you've used or heard of, and what it does.

I can... (check when you're confident)

☐ Understand cloud concepts: cloud storage vs. cloud access; sharing and collaboration via the cloud (1.6.a)

☐ Know the benefits of using cloud storage (1.6.b)

☐ Access and use the cloud: account management; upload, download, sync files between devices (1.6.c)

☐ Web apps vs. local apps: SaaS concepts; what Learning Management Systems (LMS) do (1.6.d)

☐ Web app types: online storage (Google Drive, Dropbox, OneDrive, iCloud), online office (Google apps, Office 365), database-driven CRM (Salesforce, Oracle, Workfront) (1.6.e)

UNIT 8 · SESSION 8

Security, Privacy & Safe Online Shopping

Name: _____ Date: _____

Notes Guide

1. A _____ incorporates all the parts of a good password but is easy to remember.
2. A virus is a program or piece of _____ that can erase, corrupt, or gather data.
3. After using a public network/computer, log out of any _____ and clear the browser history, cache, and _____.
4. A savvy online shopper is aware of deals that are _____.
5. A firewall _____. Antivirus _____.
6. Secure site clue: _____ and a padlock.
7. Social engineering examples: _____, _____, _____

Key Vocabulary

Term	What it means
Malware	Malicious software. The umbrella term for any software designed to harm, spy on, or take control of a device (viruses, Trojans, ransomware, spyware).
Virus	Malicious code attached to a file or program. It spreads when the infected file is opened or run, and it can erase, corrupt, or steal data.
Trojan Horse	Malware disguised as a useful or harmless program. Unlike a virus, it doesn't copy itself; it tricks you into installing it.
Ransomware	Malware that locks or encrypts your files and demands payment to unlock them.
Social Engineering	Tricking people into giving up information, money, or access instead of hacking the computer. Phishing emails, scam texts, and fake tech-support calls are examples.
Monitoring Software	Software that records what a person does on a device (websites, keystrokes, screen, location). Used by employers and parents, and also by attackers (spyware).
Antivirus Application	Software that scans for, blocks, and removes viruses and other malware. It needs regular updates to recognize new threats.
Firewall	Hardware or software that watches network traffic and blocks or allows it based on rules.
VPN	Virtual Private Network. Creates an encrypted 'tunnel' between your device and a VPN server, protecting your data on public networks or letting you securely reach a work network.
Secure Network	A network that requires a password or sign-in and encrypts traffic (for example, WPA2 or WPA3 Wi-Fi).
Unsecure Network	An open network with no password or encryption, such as some public Wi-Fi. Others nearby may be able to see your traffic.
Cookie	A small file a website saves in your browser to remember you (logins, shopping carts, settings). Some cookies track you across sites.
Cache	Copies of web page parts (images, scripts) your browser saves so pages load faster next time.
Browser History	The list of websites you have visited in a web browser.

UNIT 8: Security, Privacy & Safe Online Shopping (continued)

Practice Questions

1. Create a strong passphrase (don't use a real one!) and explain why it's strong.

2. List three red flags in scam texts or emails.

3. Write your 'public computer checklist' for when you leave a library computer.

4. What's the difference between a virus and a Trojan horse?

5. Name two things that make online shopping safer.

I can... (check when you're confident)

- ☐ Know credential management best practices: usernames; passwords and the components of a good password (1.7.a)
- ☐ Know basic account settings management (1.7.b)
- ☐ Know basic threats: viruses, Trojans, malware, social engineering (1.7.c)
- ☐ Understand the implications of monitoring software (surveillance) (1.7.d)
- ☐ Secured vs. unsecured networks (wired and wireless); public vs. private computing: browser history, cache/cookies, logging out (1.7.e)
- ☐ Know the use and importance of antivirus software; prevent virus issues; maintain/update antivirus (1.7.f)
- ☐ Know the use of firewalls and basic settings (1.7.g)
- ☐ eCommerce best practices: recognize secure connections (https); be a savvy consumer (1.7.h)
- ☐ Understand what Virtual Private Networks are (workbook skills assessment & Security video) ((VPN))

GMetrix Progress Tracker

Training Mode = learning (help on, no time limit). Testing Mode = the real-exam feel (timed, no help). Goal: 85%+ in Testing Mode before you take the certification exam.

Date	Mode (Training / Testing)	Test #	Score %	Weakest topics (from Study Guide)	Next step

My exam date: _____ Testing location: _____

My Certiport username: _____ (keep your password private!)

My Exam-Day Checklist

The week before

☐ I scored 85%+ (or at least 80% twice) in GMetrix Testing Mode

☐ I know my Certiport username and password

☐ My name in Certiport matches my ID exactly

☐ I know the exam date, time, and location

The night before

☐ Review flashcards (especially file extensions and security words)

☐ Review the step lists: Wi-Fi, power & sleep, screen saver, uninstall, Read-only, zip, Task Manager, virus scan, firewall

☐ Pack my ID and login info

☐ Get a good night's sleep

During the exam (50 minutes, about 45-50 questions)

☐ Read EVERY word: watch for NOT, BEST, FIRST

☐ About 1 minute per question. Mark hard ones and come back

☐ Cross out answers I know are wrong

☐ Don't leave anything blank

☐ Breathe. I practiced this.